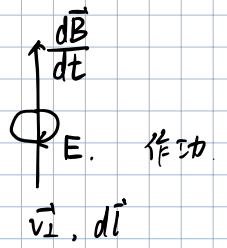
随时间缓慢变化的磁场.

磁场感生电场. $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$.



作功

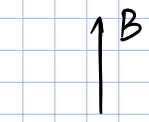
$$\frac{d(\frac{1}{2}mv_\perp^2)}{dt} = qE\frac{d\vec{l}}{dt} \qquad \delta(\tfrac{1}{2}mv_\perp^2) = \int_0^\tau qE\frac{d\vec{l}}{dt}\cdot dt$$

$$= \int_0^{\frac{2\pi}{\omega_0}} q\vec{E}\frac{d\vec{l}}{dt}\,dt$$

$$= \oint_l q\vec{E}\cdot d\vec{l}$$

$$= q\int_S (\nabla\times\vec{E})\,d\vec{S}$$

$$= -q\int_S \frac{\partial\vec{B}}{\partial t}\cdot d\vec{S}$$



正 $d\vec{S} > 0$.

负 $d\vec{S} < 0$.

$$\frac{\partial\vec{B}}{\partial t}\cdot d\vec{S} = \mp\frac{\partial B}{\partial t}dS$$

$$\frac{\partial B}{\partial t} = \frac{\delta B}{T} = \frac{\omega_c}{2\pi}\delta B.$$

$$\delta(\tfrac{1}{2}mv_\perp^2) = +q\pi r_\perp^2\frac{\partial B}{\partial t} \qquad \pm:\begin{cases} +: & q>0 \\ -: & q<0. \end{cases}$$

$$= \pm q\pi\frac{v_\perp^2}{\omega_c}\frac{m}{|q|B}\frac{\partial B}{\partial t}$$

$$= \underbrace{\frac{\frac{1}{2}mv_\perp^2}{B}}_{\mu}\underbrace{\frac{2\pi}{\omega_c}}_{T}\frac{\partial B}{\partial t} = \mu\delta B.$$

$$\delta(\mu B) = \mu\delta B \Rightarrow \delta\mu = 0, \quad \mu = const.$$

磁通 $\phi = BS = B\pi r_\perp^2 = B\pi V_\perp^2/\omega_c^2 = B\pi V_\perp^2\frac{m}{q^2B^2} = \mu\frac{2\pi m}{q^2}$

$\frac{1}{\omega_c} = \frac{m}{qB}$

守恒.

$$r_L^2 = \frac{v_\perp^2}{\omega_c^2} = \frac{v_\perp^2 m^2}{q^2 B^2} = \mu \frac{2m}{q^2 B^2} \qquad r_L^2 \propto \mu B^{-}, \; B \uparrow.$$